



LINEAR ALGEBRA

UE19MA251

APARNA B S

Department of Science and Humanities

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Agenda



- Singular value and singular vector
- Positivity of a Singular value
- Number of Singular values
- Singular value decomposition(SVD)
- Proofs of SVD
- Matrices and SVD
- Singular vectors and fundamental subspaces

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Singular Value and Singular Vector

A **singular value** of a real matrix A is the square root of a non-zero eigenvalue of $(A^T A)$.

It means to find the singular values of A , one needs to find the non-zero eigenvalues of $(A^T A)$.

Singular vector. If σ is a singular value of A , then there exists $v \neq 0$ such that

$$(A^T A) v = \sigma^2 v$$

Such a v is called a right singular vector of A with singular value σ . It is an eigenvector of $(A^T A)$ with eigenvalue σ^2 .

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Positivity of a Singular Value

A singular value is always positive.

The matrix $(\mathbf{A}^T \mathbf{A})$ is positive semi-definite:

$$\mathbf{x}^T (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{A}\mathbf{x})^T (\mathbf{A}\mathbf{x}) = \|\mathbf{A}\mathbf{x}\|^2 \geq 0$$

so the eigenvalues of $(\mathbf{A}^T \mathbf{A})$ must be non-negative, and the non-zero eigenvalues must be positive. Hence a singular value is positive.

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Number of Singular Values



A matrix of **rank** r has exactly r singular values.

Proof. Note $(A^T A)x = 0 \Leftrightarrow x^T A^T A x = 0 \Leftrightarrow Ax = 0$, so $\mathcal{N}(A^T A) = \mathcal{N}(A)$. Let A be of order $m \times n$ with rank r . Then $\dim \mathcal{N}(A) = n - r = \dim \mathcal{N}(A^T A)$. Matrix $(A^T A)$ is non-defective, so the algebraic multiplicity of eigenvalue 0 is $(n - r)$. It follows that the total algebraic multiplicities of the non-zero eigenvalues of $(A^T A)$ is

$$n - (n - r) = r$$

Notation. Singular values are denoted by $\sigma_1, \dots, \sigma_r$.

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Singular Value Decomposition (SVD)

A real matrix can be decomposed by its singular values and singular vectors. This is called singular value decomposition.

A matrix of order $m \times n$ has SVD

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

where $\mathbf{\Sigma}$ is an $m \times n$ "diagonal" matrix with the singular values of $\mathbf{A}$ as the leading diagonal elements, $\mathbf{U}$ is an $m \times m$ orthogonal matrix with the eigenvectors of $(\mathbf{A}\mathbf{A}^T)$ as columns, and $\mathbf{V}$ is an $n \times n$ orthogonal matrix with the eigenvectors of $(\mathbf{A}^T\mathbf{A})$ as columns.

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Proof of SVD



Let r be the rank of A . Let $\sigma_1 \dots \sigma_r$ be the singular values of A . Let $v_1 \dots v_r$ be orthonormal eigenvectors of $(A^T A)$ with positive eigenvalues σ_i^2 and $u_1 \dots u_r$ be defined by $u_i = \frac{Av_i}{\sigma_i}$. Note

$$(AA^T)u_i = \frac{AA^T Av_i}{\sigma_i} = \frac{A\sigma_i^2 v_i}{\sigma_i} = \sigma_i^2 u_i, \quad i = 1, \dots, r$$

So u_i is an eigenvector of (AA^T) with the same eigenvalue σ_i^2 . Let $v_{r+1} \dots v_n$ be orthonormal eigenvectors of $(A^T A)$ with eigenvalue 0, and $u_{r+1} \dots u_m$ be eigenvectors of (AA^T) with eigenvalue 0. Construct matrices U and V by

$$U = \begin{bmatrix} u_1 & \dots & u_m \end{bmatrix}, \quad V = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}$$



THANK YOU

Aparna B. S

Department of Science & Humanities

aparnabs@pes.edu